

AI-Powered Patient Case-Taking Software

Pre-Consultation Clinical Intelligence & Multilingual Medical Document Digitization Platform

- ◆ **Voice-First Intake:** Natural spoken conversation in Hindi & English eliminates complicated forms.
- ◆ **Adaptive Follow-Up:** Context-aware AI generates dynamic medical questions based on symptoms.
- ◆ **Document OCR Pipeline:** Extracts medicines, dosages & lab values from paper prescriptions & reports.
- ◆ **Physician-Ready Report:** Structured, standardized clinical case summary generated prior to OPD visit.



CORE PARADIGM: Patient Speaks → AI Structures → History Verified → Physician Decides

The Clinical History-Taking Bottleneck in Hospital OPDs

Why conventional pre-consultation workflows overburden doctors and delay patient care

[CHALLENGE 1] Severe OPD Overcrowding & Time Drain

- **Repetitive questioning:** Doctors spend 50–60% of consultation time collecting basic demographic and symptom data.
- **Rushed physical exams:** Leaves inadequate time for in-depth physical examination and clinical counseling.
- **Physician burnout:** High patient volume causes severe cognitive exhaustion and risks diagnostic oversight.

[CHALLENGE 2] Fragmented Paper Records & Lost Medical History

- **Disorganized records:** Patients arrive with crumpled prescriptions, torn discharge summaries, and lost lab slips.
- **Critical data omission:** Vital drug allergies, prior surgical histories, and current chronic medications are often missed.
- **Redundant lab testing:** Inability to retrieve prior test records leads to costly duplicate investigations.

[CHALLENGE 3] Linguistic & Communication Barriers

- **Symptom articulation:** Patients struggle to articulate clinical onset, timeline, and progression accurately.
- **Lack of vernacular intake:** Absence of automated Hindi & vernacular pre-intake causes diagnostic blind spots.
- **Intimidating portals:** Conventional digital hospital systems force users through complex, confusing forms.



THE CLINICAL REALITY IN NUMBERS

- ▶ **Current OPD Reality:** Average 10–12 minutes spent taking basic history; only 3–4 minutes left for clinical diagnosis.
- ▶ **With Our Solution:** Comprehensive physician-ready clinical summary ready in 30 seconds before patient enters.

AI-Assisted Digital Case Taking Before Doctor Consultation

Transforming patient speech and paper records into structured physician-ready documentation

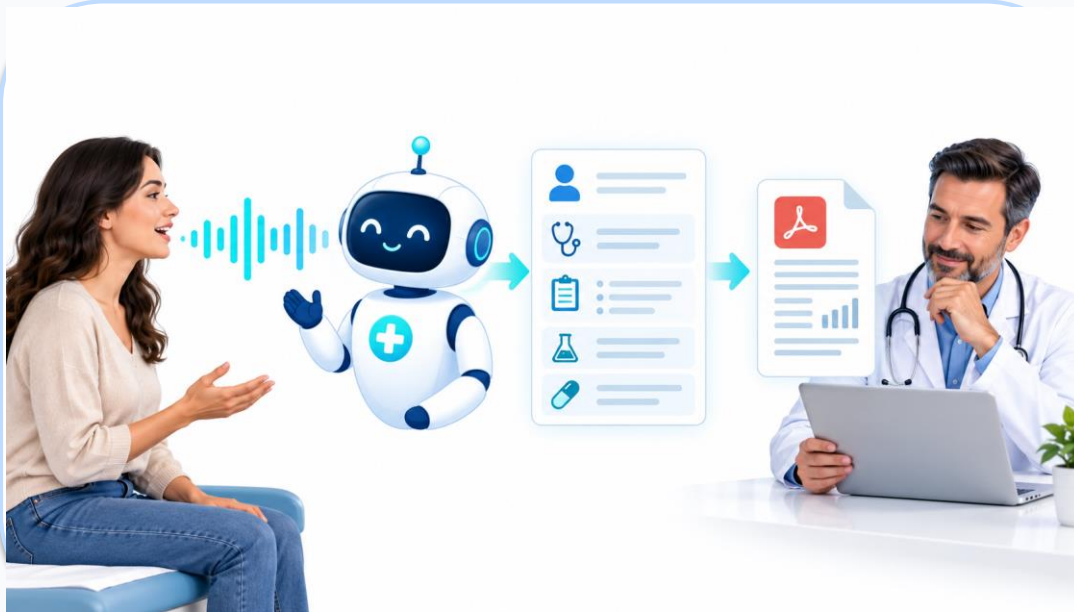
1. Patient Arrival & Registration

2. Bilingual Voice Interview

3. Document OCR Digitization

4. Patient Review & Confirmation

5. Physician Consultation



[PILLAR 1] Natural Spoken Dialogue

- **Conversational Intake:** Patients speak freely in Hindi or English without navigating complex dropdowns.
- **Universal Inclusivity:** Specially designed for rural and elderly citizens who find digital forms intimidating.

[PILLAR 2] Adaptive Clinical Inquiries

- **Context-Driven Logic:** AI asks intelligent follow-up questions regarding duration, severity, and progression.
- **Non-Linear Exploration:** Dynamically explores prior medications taken and patient therapeutic response.

[PILLAR 3] Patient-in-the-Loop Review

- **Explicit Confirmation:** Patient reviews structured summary on screen and corrects any errors before finalization.
- **High-Fidelity PDF Delivery:** A standardized clinical report is automatically ready for doctor review.

CLINICAL SAFETY MANDATE: The AI system is strictly designed for clinical case-taking, information structuring, and document digitization. The AI NEVER performs autonomous diagnosis, never issues prescriptions, and never decides medical treatment. The licensed physician remains the sole clinical decision-maker.

The Complete 9-Module Healthcare Ecosystem

An integrated architecture spanning patient intake, AI consultation, hospital workflows, and governance



[CLINICAL CORE] Pillar 1: Patient Intake & Clinical Core

- **Module 1 — Patient Registration:** Frictionless demographic intake, vitals (height/weight), and existing disease checklist.
- **Module 2 — AI Consultation:** Voice assistant, adaptive questioning, document OCR, and patient review.
- **Module 3 — My Report History:** Searchable, immutable longitudinal repository of previous consultation reports.

[ACCESSIBILITY] Pillar 2: Patient Continuity & Accessibility

- **Module 4 — Notification Center:** Real-time alerts for appointments, medicine schedules, and lab reports.
- **Module 5 — My Profile & Settings:** Personal details, profile photo, notification preferences, and privacy controls.
- **Module 6 — Multi-Language Support:** Seamless Hindi & English bilingual UI, voice prompts, and speech recognition.
- **Module 7 — Help & Support:** User guides, FAQs, feedback, and technical issue reporting.

[HOSPITAL OPS] Pillar 3: Hospital Operations & Governance

- **Module 8 — Hospital Services:** Doctor appointment booking, hospital pharmacy, lab test booking, and ambulance requests.
- **Module 9 — Hospital Admin Portal:** 7-tier RBAC, operational CRUD for doctors/departments, analytics, and audit logging.

STRICT ARCHITECTURAL SCOPE: Exactly 9 Approved Modules operating in unified cohesion across 43 development phases. Zero Module 10.

Module 1: Frictionless Patient Registration

Pre-consultation onboarding capturing vital demographics, baseline health, and existing conditions

[WORKFLOW] 7-Step Streamlined Workflow

- **End-to-End Flow:** Landing → Basic Details → Personal Info → Address & Medical Info → Mobile OTP → Account Ready → Dashboard.
- **Rapid Intake:** Designed to take under 2 minutes for first-time hospital visitors.

[DATA MODEL] Essential Demographic & Health Vitals

- **Demographics:** Captures full name, date of birth (automatic age calculation), gender, mobile, email, and address.
- **Physical Vitals:** Collects height (cm), weight (kg), and optional blood group.

[CLINICAL INPUT] Pre-Existing Conditions Checklist

- **Rapid Screening:** Predefined checkboxes for Diabetes, Hypertension, Asthma, Thyroid Disorder, Heart Disease, Other, None.
- **AI Seeding:** Selected conditions directly seed the AI Consultation for relevant follow-up inquiry.

[DESIGN RULE] Clean Architectural Separation

- **Lean Registration:** Symptoms and detailed medical history are deferred to Module 2 (AI Consultation).
- **Profile Photo Decoupled:** Profile picture upload is managed later in Module 5 Settings.

CarePlus
Your Health Partner

Home About Services Contact English Login

Module 1 – Patient Registration
From First Visit to Your Personalized Health Journey

Module 01

Better Healthcare Starts with You
Register now and take the first step towards a healthier tomorrow.
Register Now →
Already have an account? Login

Create Your Account
Please fill in your details to register

Full Name
Ravi Kumar

Mobile Number
9876543210

Email Address
ravi@example.com

Password

Continue →
Already have an account? Login

Personal Information
Tell us more about yourself

Date of Birth
15-08-1990

Gender
Male

Blood Group
B+

Height (kg)
70

Continue →
← Back

Address & Medical Information

Address
123, Green Park, New Delhi, Delhi

Existing Diseases (if any)
Diabetes × Hypertension ×
Type and press enter

I agree to the Terms & Conditions and Privacy Policy

Create Account →
← Back

1 Landing Page
Click on "Register Now" to create your account

2 Basic Registration
Enter your basic details

3 Personal Information
Provide your personal and health details

4 Address & Health Details
Add your address and existing medical conditions

5 OTP Verification
Enter the OTP sent to your mobile number

6 Registration Successful
Your account is ready

7 User Dashboard
Access all features after successful registration

Verify Your Mobile Number
We have sent a 6-digit OTP to +91 9876543210
4 8 2 9 1 6
Didn't receive OTP? Resend (00:30)
Verify →

Account Created Successfully!
Welcome to CarePlus. Your account has been created successfully.
Go to Dashboard →

Hello, Ravi Kumar
Welcome to your health dashboard

Home
AI Consultation Start now
My Reports
Appointments
Notifications 2
My Profile
Help & Support
Logout

Book Appointment Find a doctor
Lab Tests Book a test
Pharmacy Order medicines

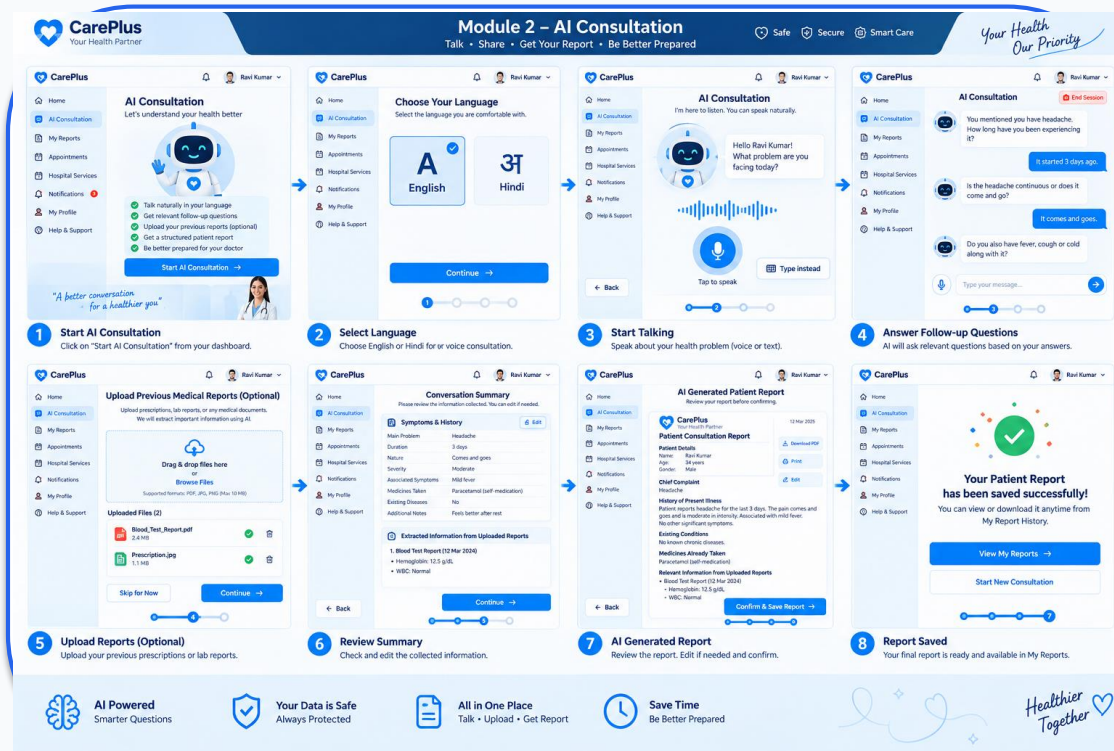
Your Health Journey Starts Here
Use our AI-powered consultation to prepare for your doctor visit.

Your Profile
View & Edit

Stay Healthy Stay Informed
Your health is our priority.

Module 2: Voice-First AI Consultation & Case Taking

Dynamic conversational engine powered by speech recognition and context-driven clinical inquiries



[VOICE ARCHITECTURE] Bilingual Voice Processing Pipeline

- **Speech-to-Text (STT):** Speech audio processed via faster-whisper local engine for low latency in Hindi & English.
- **Context & Intent Engine:** Extracts chief complaints, symptom timelines, and anatomical references from conversational speech.
- **Text-to-Speech (TTS):** Synthesizes natural voice responses to ask follow-up questions, creating a comfortable experience.

[CLINICAL CONVERSATION] Adaptive Context-Driven Questioning

- **Open Chief Complaint:** Begins naturally: 'What problem are you facing today?' without suggesting illnesses.
- **Dynamic Medical Drilldown:** Inquires into duration ('How long?'), progression ('Continuous or intermittent?'), and severity.
- **Medication History:** Checks for self-medication (e.g., Paracetamol) and therapeutic response.

[SAFETY VERIFICATION] Patient Review & Edit Mechanism

- **Pre-Finalization Summary:** Extracted symptoms, duration, and past medications are displayed in an editable summary.
- **Correction Interface:** Patient can modify any misunderstood information before confirming the case report.
- **Immutable Finalization:** Once confirmed, the system compiles the structured report for physician review.

Medical Document Intelligence & OCR Pipeline

Extracting structured clinical data from paper prescriptions, laboratory tests, and discharge summaries



[INGESTION] Stage 1: Multi-Format Document Ingestion

- **Supported Formats:** Accepts PDF, JPG, and PNG uploads of physical prescriptions, lab slips, and discharge summaries.
- **Private Storage:** Encrypted document storage in Supabase private buckets accessible only via signed URLs.

[PREPROCESSING] Stage 2: Image Preprocessing Pipeline

- **OpenCV & Pillow:** Applies automatic deskewing, noise reduction, shadow removal, and contrast enhancement.
- **Adaptive Binarization:** Optimizes handwritten and faded clinical text for high OCR recognition accuracy.

[OCR ENGINES] Stage 3: Dual-Engine OCR Recognition

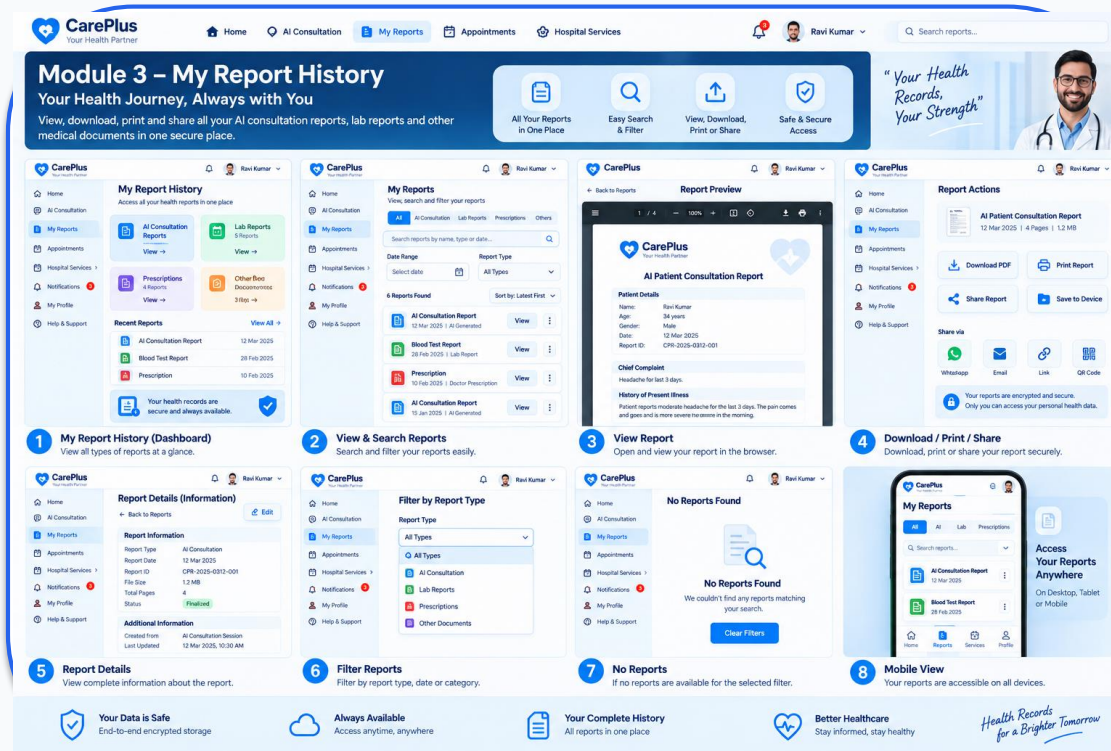
- **PyMuPDF Engine:** Instantly parses native digital text and embedded metadata from digital laboratory PDFs.
- **PaddleOCR Engine:** Excels at complex multi-column document layouts, Hindi/English text, and handwritten notes.

[SYNTHESIS] Stage 4: Structured Clinical Entity Extraction

- **Medication Parsing:** Extracts drug names, dosages (e.g., 500mg), frequency, and treatment duration.
- **Diagnostic Synthesis:** Identifies abnormal lab values (Hb, glucose), past surgeries, and merges with AI context.

Module 3: Structured Patient Case Report & History

Physician-ready clinical documentation with immutable, patient-accessible longitudinal history



[REPORT SCHEMA] Standardized Clinical Report Schema

- **Clinical Sections:** Patient Demographics, Chief Complaint, History of Present Illness (HPI), Existing Conditions.
- **Records Integration:** Prior medications taken, response to drugs, and extracted laboratory data summarized.

[EXPORT ENGINE] High-Fidelity PDF Generation Engine

- **Jinja2 + WeasyPrint:** HTML/CSS semantic templates compiled into clean, printable A4 clinical PDF documents.
- **Instant Availability:** Available for in-app viewing, offline PDF download, direct printing, or QR code sharing.

[MANAGEMENT] Search & Intelligent Filtering

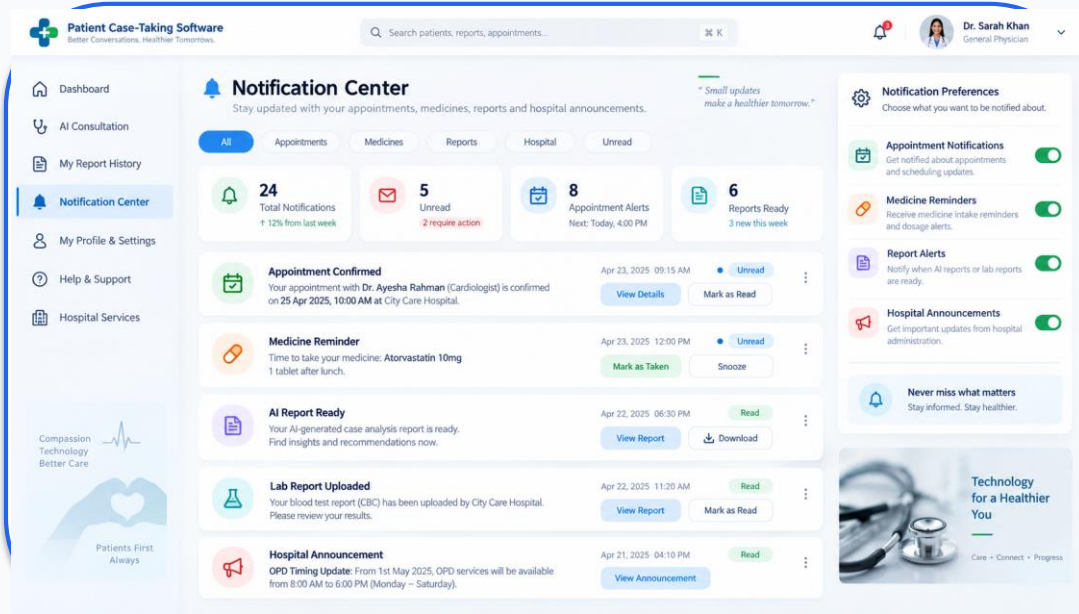
- **Multi-Criteria Search:** Filter by date range, consultation type, or doctor name in seconds.
- **Category Segregation:** Separate tabs for AI Consultation Reports, Lab Reports, Prescriptions, and External Documents.

[DATA INTEGRITY] Write-Once Record Protection

- **Immutable Final Reports:** Patients cannot delete finalized medical records, preserving legal & clinical integrity.
- **Administrative Governance:** Record modifications restricted to authorized administrators under strict audit logging.

Patient Continuity, Accessibility & Support Infrastructure

Ensuring holistic engagement, bilingual accessibility, account control, and patient guidance



[MODULE 4] Module 4 — Notification Center

- **Real-Time Multi-Alerts:** Instant alerts for appointment bookings, scheduled medicine reminders, and lab report readiness.
- **Hospital Circulars:** Official announcements regarding OPD timings, department notices, and health camps.

[MODULE 5] Module 5 — My Profile & Settings

- **Profile Management:** View and update permitted personal details, contact info, and emergency contacts.
- **Privacy & Photo Controls:** Profile photo management and granular patient consent toggles for medical data sharing.

[MODULE 6] Module 6 — Multi-Language Support

- **Hindi & English Interface:** Full UI localization via i18next across all navigation, forms, and clinical summaries.
- **Bilingual Voice Engine:** Configurable speech recognition and voice playback in Hindi or English with auto-persistence.

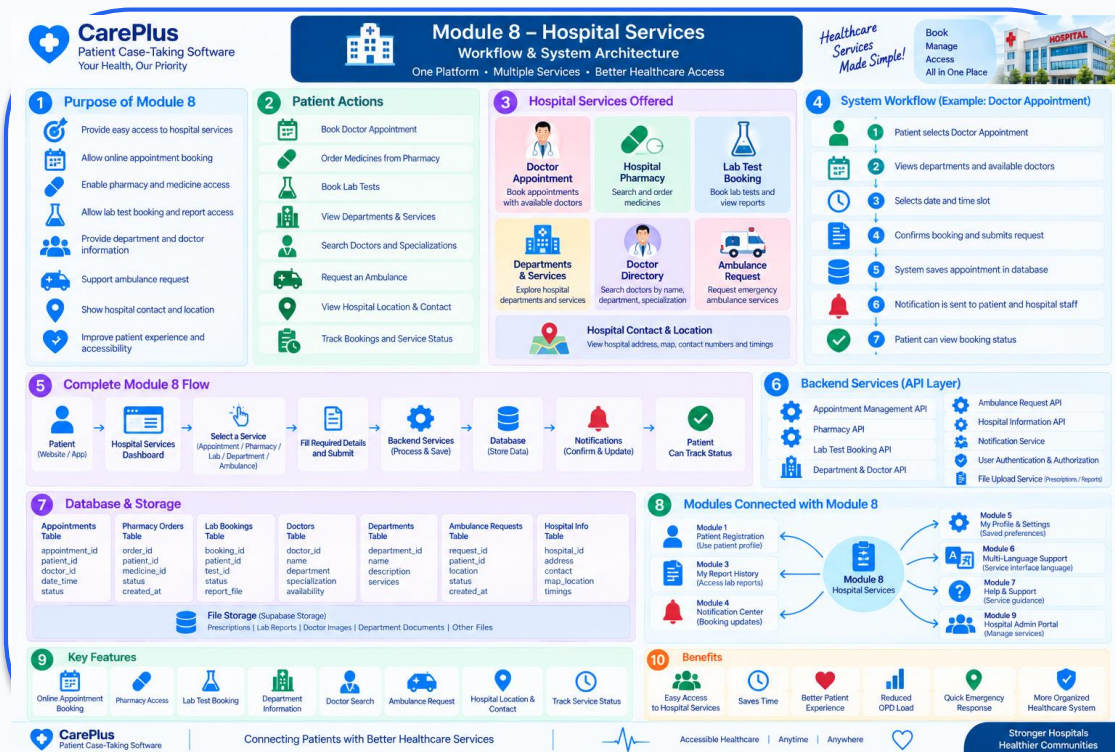
[MODULE 7] Module 7 — Help & Support

- **Interactive Guidance:** Illustrated visual user guides, searchable FAQs, and patient experience feedback ratings.
- **Issue Reporting:** Direct bug reporting with screenshot upload capability to hospital technical support.

PATIENT CONTINUITY HIGHLIGHT: Modules 4 to 7 deliver holistic patient care—keeping citizens informed, respecting privacy, removing language barriers, and guiding first-time users.

Module 8: Integrated Hospital Services Ecosystem

Extending patient care with appointment booking, pharmacy access, lab diagnostics, and emergency care



[APPOINTMENTS] Doctor Appointment Booking

- **Multi-Department Directory:** Search specialists by clinical department, OPD timings, and consultation fees.
- **Real-Time Slot Validation:** Interactive date/slot selection with database concurrency locking to prevent overbooking.

[PHARMACY] Hospital Pharmacy Services

- **Medicine Inventory Search:** Real-time stock checking for prescribed drugs and essential Ayush medicines.
- **Counter Pickup or Delivery:** Select hospital pharmacy counter pickup or address delivery where supported.

[LAB TESTS] Diagnostic Lab Test Booking

- **Diagnostic Test Catalogue:** Explore available pathology and radiology tests with clear preparation instructions.
- **Automated Report Delivery:** Completed lab reports upload directly into the patient's Module 3 history.

[EMERGENCY CARE] Ambulance & Emergency Access

- **Instant Dispatch Request:** One-tap emergency request capturing patient contact and GPS pickup location.
- **Safety Bypass Rule:** EMERGENCY CASES DIRECTLY ACCESS HOSPITAL CARE—NEVER FORCED THROUGH AI CONSULTATION.

Module 9: Hospital Admin Portal & Operational Governance

A centralized, authenticated console for hospital operations, resource rosters, and compliance monitoring



[SECURITY RBAC] 7-Tier Role-Based Access Control (RBAC)

- Super Admin:** Full system configuration, user provisioning, and role assignment.
- Clinical Staff Roles:** Appointment Staff, Pharmacy Staff, Lab Technicians, and Emergency Dispatchers.
- Department & Audit Roles:** Department Managers and View-Only Medical Auditors with restricted data scope.

[HOSPITAL CRUD] Comprehensive Administrative CRUD Operations

- Staff & Doctor Roster:** Create, update, and manage doctor availability, OPD time slots, and departmental assignments.
- Pharmacy & Inventory:** Monitor medicine stock levels, update pricing, and track dispensation.
- Lab Test Management:** Configure diagnostic test catalogues, upload patient lab reports, and manage booking queues.

[AUDIT & KPI] Operational Analytics & Security Audits

- Live KPI Dashboard:** Real-time metrics: Total Registered Patients, Daily Appointments, Active Consultations, Lab Volume.
- Emergency Queue Monitor:** Live tracking of ambulance requests and emergency resource allocation.
- Immutable Audit Logging:** Every administrative access, data modification, and report view is logged for compliance.

Decoupled Microservice Architecture & Technology Stack

Enterprise-grade, open-source stack designed for high throughput, data privacy, and clinical reliability

[TIER 1 • CLIENT] Frontend Presentation Layer

- **Framework:** React 18 Single Page Application (SPA) built with Vite for sub-second hot reload.
- **Language:** TypeScript providing static type contracts across all patient and admin interfaces.
- **Design System:** Tailwind CSS paired with accessible shadcn/ui primitives and Lucide icons.
- **State & Forms:** TanStack Query for server cache, React Hook Form with Zod validation schemas.
- **Localization:** i18next bilingual engine enabling seamless real-time Hindi & English switching.
- **Accessibility:** WCAG 2.1 AA compliant UI, mobile-first responsive layout, and high-contrast theme.

CORE TECHNOLOGIES & LIBRARIES

React 18

TypeScript

Vite SPA

Tailwind CSS

shadcn / UI

i18next

[TIER 2 • API & AI] Application API & AI Services

- **API Framework:** Python FastAPI providing high-throughput asynchronous REST endpoints and OpenAPI docs.
- **Data Validation:** Pydantic v2 schemas ensuring strict request/response validation and type safety.
- **Speech Pipeline:** faster-whisper self-hostable STT pipeline for low-latency Hindi & English transcription.
- **Document OCR:** PaddleOCR + PyMuPDF + OpenCV for document deskewing, table detection, and extraction.
- **Clinical PDF:** Jinja2 semantic templating compiled to A4 clinical PDFs via WeasyPrint CSS-to-PDF.
- **Async Workers:** Non-blocking background task workers ensuring sub-second API responsiveness.

CORE AI & BACKEND ENGINES

Python 3.11

FastAPI

Whisper AI

PaddleOCR

WeasyPrint

OpenCV

[TIER 3 • DATA & STORAGE] Cloud Platform & Storage

- **Relational Engine:** Supabase PostgreSQL acting as the central transactional database for all 9 modules.
- **Identity & Auth:** Supabase Auth issuing cryptographically signed JWTs for patients and administrators.
- **Row Level Security:** PostgreSQL RLS enforcing data isolation so patients access strictly their own records.
- **Private Storage:** Encrypted S3-compatible private buckets; documents accessible only via time-limited signed URLs.
- **Realtime & Cron:** Supabase Realtime for instant alerts and pg_cron for scheduled medicine reminders.
- **Data Durability:** Automated daily point-in-time recovery (PITR) backups and AES-256 encryption at rest.

DATABASE & SECURITY INFRASTRUCTURE

PostgreSQL

Supabase Auth

pg_cron

Postgres RLS

AES-256 S3

JWT Claims

OPEN-SOURCE SOVEREIGNTY: 100% open-source baseline. Zero proprietary cloud lock-in. Can be self-hosted on hospital premise servers in full compliance with Indian data sovereignty.

Privacy, Data Security & Clinical Safety Guardrails

Built-in compliance with healthcare data confidentiality, patient consent, and physician primacy

[DATA ACCESS] PostgreSQL Row Level Security (RLS)

- **Database-Level Isolation:** RLS policies guarantee that patients can query ONLY their personal medical records.
- **Cryptographic JWT Claims:** Every FastAPI request validates Supabase Auth JWT identity before database execution.
- **Prevention of Cross-Tenant Leaks:** Even if frontend API calls are manipulated, the database layer strictly blocks unauthorized queries.

[FILE SECURITY] Private Storage & Signed URLs

- **Encrypted Document Vault:** All uploaded prescriptions, blood test reports, and generated case PDFs reside in private buckets.
- **Time-Limited Access:** Documents are accessed strictly through signed, expirable URLs (e.g., 15-minute expiration).
- **Zero Public Bucket Exposure:** No health record is ever exposed via unauthenticated public URLs.

[COMPLIANCE] Patient Consent & Audit Trails

- **Granular Consent Architecture:** Patients explicitly authorize medical record sharing with specific hospital departments.
- **Immutable Administrative Audit Logs:** Every administrative action, record view, and role modification is permanently logged with timestamp.
- **DPDP Act 2023 Compliance:** Designed in alignment with India's Digital Personal Data Protection Act standards.

[CLINICAL SAFETY] Clinical Safety & Non-Diagnostic Boundary

- **Zero Autonomous Diagnosis:** The AI system explicitly refrains from providing medical diagnoses or differential illness probabilities.
- **Zero Prescription Authority:** The software cannot prescribe medication, adjust dosages, or modify existing treatment plans.
- **Doctor in Complete Control:** The physician conducts physical examination, applies clinical judgment, and makes all final medical decisions.

SIH Evaluation Alignment & Tangible Healthcare Impact

Demonstrating high technical novelty, practical feasibility, and massive national healthcare impact

[CRITERION 1] Novelty & Innovation

- **Bilingual Conversational Intake:** First AI case-taking engine tailored for Indian hospital OPDs with natural Hindi and English speech.
- **Document Intelligence Fusion:** Blends voice symptom narrative with OCR extraction from prior prescriptions into a single coherent summary.

[EFFICIENCY] 50–60% Reduction in History-Taking Time

- **From 12 min to 30 sec:** Doctors review a pre-compiled, structured case summary in 30 seconds rather than spending 10 minutes typing.
- **Higher OPD Throughput:** Clinics can manage heavy patient loads efficiently without sacrificing consultation quality.

[CRITERION 2] Feasibility & Practicability

- **Lightweight Open-Source Stack:** Operates efficiently on standard commodity hardware without requiring million-dollar GPU infrastructure.
- **Rapid Hospital Onboarding:** Modular deployment enables easy adoption by small Ayurvedic clinics or massive tertiary medical colleges.

[ACCURACY] Zero Overlooked Medical History

- **Comprehensive HPI Coverage:** Systematically captures symptom onset, progression, severity, and prior drug reactions without human error.
- **No More Duplicate Tests:** Digitized past lab reports prevent patients from repeating expensive blood tests unnecessarily.

[CRITERION 3] Scalability & Sustainability

- **Multi-Tenant Architecture:** Designed to serve multi-department hospitals with horizontal scaling across outpatient departments.
- **Low Operating Cost:** Local faster-whisper and PaddleOCR minimize costly recurring cloud API subscriptions.

[CLINICAL VALUE] Empowered Doctor-Patient Interaction

- **Restoring Human Connection:** Physicians spend consultation time listening, physically examining, and explaining treatments.
- **Reduced Doctor Burnout:** Eliminates repetitive administrative typing, allowing doctors to focus on clinical excellence.

Demonstration-Ready Vision & ABDM Integration Roadmap

From hackathon prototype to nationwide healthcare infrastructure for the Ministry of Ayush



PHASE 1: DELIVERED HACKATHON PROTOTYPE (Current Status)

- **Production Scope:** Complete 9-module working platform: Registration, AI Voice Consultation, OCR, Reports, Admin Portal.
- **Bilingual Readiness:** Fully functional Hindi and English speech-to-text, adaptive questioning, and PDF generation.

PHASE 2: ABDM & ABHA INTEGRATION (3–6 Months Roadmap)

- **Ayushman Bharat Digital Mission:** ABHA ID creation and linking for seamless digital health record exchange.
- **FHIR Interoperability:** FHIR-compliant structured case report export for integration with hospital EHRs and e-Sanjeevani.

PHASE 3: NATIONAL INSTITUTIONAL ROLLOUT (6–12 Months)

- **Clinical Pilot Deployment:** Pilot deployment across OPDs at All India Institute of Ayurveda (AIIA) and partner Ayush hospitals.
- **Regional Indian Languages:** Expanding voice models to 12+ regional Indian languages to bridge healthcare accessibility nationwide.

“Empowering physicians with structured clinical intelligence, giving every patient a natural voice in their care.”